

RAG Pipeline Evaluation Report

Performance Assessment of Retrieval-Augmented Generation Architectures

PREPARED BY / AUTHOR

Massna Ijaz

PROJECT MENTOR

Anzair

1. Project Overview

This evaluation report assesses the performance of the Retrieval-Augmented Generation (RAG) pipeline for the **Academic Research Assistant**. The primary objective is to objectively analyze and compare three distinct retrieval modes—**Dense**, **Hybrid**, and **Hybrid + Reranking**—utilizing a golden dataset of 20 predefined evaluation queries across specialized academic literature.

RAG System Architecture

- **Vector Database:** Qdrant Cloud
- **Embedding Model:** Voyage AI
- **LLM Engine:** Groq (Llama-3.1-8b-instant)
- **Frontend Framework:** Next.js (React)
- **Backend API:** FastAPI (Python)

Experimental Configuration

- **Chunk Size:** 700 tokens
- **Chunk Overlap:** 120 tokens
- **Top-K Retrieval:** 7 chunks
- **Rerank Top-N:** 50 candidates
- **LLM Temperature:** 0.3 (Deterministic)

Dataset Information & Scope

The evaluation corpus comprises **4 academic PDFs** covering foundational artificial intelligence domains: *Large Language Models*, *Retrieval-Augmented Generation*, *AI Agents*, and *Fine-Tuning & PEFT*.

Total Corpus Chunks: 85 | **Evaluation Queries:** 20 golden benchmark queries.

2. Retrieval Mode Performance Results

The tables below detail the individual quantitative performance metrics for each retrieval approach evaluated across the 20 golden benchmark queries.

Dense Retrieval

Metric	Value
Precision@7	0.72
Recall@7	0.65
MRR	0.78
nDCG	0.75
Hit Rate	0.85
Avg. Retrieval Latency	10,744.34 ms
Avg. Generation Latency	578.47 ms
Avg. Total Latency	11,322.82 ms

Hybrid Retrieval

Metric	Value
Precision@7	0.81
Recall@7	0.76
MRR	0.85
nDCG	0.83
Hit Rate	0.95
Avg. Retrieval Latency	17,331.22 ms
Avg. Generation Latency	810.17 ms
Avg. Total Latency	18,141.40 ms

Hybrid + Rerank Retrieval Optimal Accuracy

Metric	Value
Precision@7	0.88
Recall@7	0.85
MRR	0.92
nDCG	0.90
Hit Rate	0.98
Avg. Retrieval Latency	15,089.89 ms
Avg. Generation Latency	591.28 ms
Avg. Total Latency	15,681.16 ms

3. Comprehensive Comparison Table

A consolidated side-by-side comparison illustrating quality metrics, latency breakdown, and system hyperparameters across all three retrieval architectures.

Retrieval Mode	Prec.@7	Recall@7	MRR	nDCG	Hit Rate	Retrieval Lat. (ms)	Gen. Lat. (ms)	Total Lat. (ms)	Top-K	Chunk Size	Overlap	Queries
Dense	0.72	0.65	0.78	0.75	0.85	10,744.34	578.47	11,322.82	7	700	120	20
Hybrid	0.81	0.76	0.85	0.83	0.95	17,331.22	810.17	18,141.40	7	700	120	20
Hybrid + Rerank	0.88	0.85	0.92	0.90	0.98	15,089.89	591.28	15,681.16	7	700	120	20

4. Technical & Architectural Analysis

Latency Analysis & API Bottlenecks

The latency measurements reveal that Dense retrieval is the fastest approach, averaging 10,744.34 ms for retrieval. Hybrid search is significantly slower at 17,331.22 ms, while Hybrid + Rerank averaged 15,089.89 ms. This elevated latency across all retrieval methods is primarily driven by the **Voyage AI API rate limit**, which enforces a mandatory 65-second sleep cycle when encountering the 3 RPM (Requests Per Minute) free-tier limit, thereby skewing the overall averages. Conversely, LLM generation latency remains exceptionally fast and consistent (between 578 ms and 810 ms) due to Groq's high-speed inference engine.

Chunk Size & Overlap Strategy

A **Chunk Size of 700 tokens** was utilized across the pipeline. Given that the entire corpus consists of 85 chunks across 4 comprehensive PDFs, this larger chunk size successfully preserves deep semantic context within each individual passage. This depth is highly beneficial for conceptual academic queries that require synthesizing complex ideas. Furthermore, the **Chunk Overlap of 120 tokens** ensures semantic continuity between adjacent passages, mitigating the risk of truncating critical explanations or sentences mid-thought and ensuring that the Groq LLM consistently receives complete context.

Top-K Window & Context Surface Area

With Top-K configured to 7, retrieving 7 chunks of 700 tokens generates a substantial context window of approximately **4,900 tokens** passed directly into the LLM prompt. This broad surface area maximizes the probability that the LLM has access to the exact factual evidence required to formulate an accurate answer, balancing computational retrieval load with generation thoroughness.

Retrieval Strategy Evaluation

Hybrid + Reranking significantly outperforms the other modes across all qualitative benchmarks. It achieved the highest Precision@7 (0.88) and an exceptional Hit Rate (0.98), demonstrating that cross-

encoder reranking effectively evaluates subtle semantic nuances to surface the most relevant chunks to the very top positions. Dense retrieval performed the lowest because pure vector embeddings occasionally miss exact technical terminology and keyword matches present in academic literature. Standard Hybrid search bridges this gap by incorporating BM25 sparse keyword matching, while Reranking provides the vital finishing step to ensure the final ordering is semantically optimal.

5. Key Findings & Conclusion

Summary of Final Findings

- ✓ **Quality Metrics Dominance:** Hybrid + Reranking is the undisputed winner for accuracy and relevance, achieving an impressive **0.90 nDCG score** and a **0.98 Hit Rate**. Dense retrieval falls behind at 0.75 nDCG due to missing exact academic terminology matches.
- ✓ **Rate Limit Latency Impact:** The average retrieval latency figures are heavily skewed by the Voyage AI free-tier limit (3 requests per minute), which introduces a mandatory 65-second block during concurrent query processing. In a production environment with enterprise tier limits, retrieval latency would drop by orders of magnitude.
- ✓ **High-Speed LLM Generation:** Groq's Llama-3.1-8b-instant inference engine is exceptionally rapid and reliable, averaging well under 1 second (578 ms – 810 ms) to synthesize comprehensive, well-structured academic responses from large context windows.

Final Recommendation & Conclusion

The RAG pipeline evaluation demonstrates a classic trade-off between retrieval speed and accuracy. While Dense retrieval offers slightly faster median response times, Hybrid + Reranking provides vastly superior retrieval precision (**0.88 vs. 0.72 Precision@7**). For an **Academic Research Assistant** where factual accuracy, comprehensiveness, and precision are paramount, **Hybrid + Rerank is the unequivocally recommended retrieval mode**, easily justifying the slight computational overhead.